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Predicting customer purchase using logistic regression

Dataset

Customer	Time on site (x_1)	Pages viewed (x_2)	Purchase (y)
A	1	9	0
B	2	3	0
C	3	7	1
D	5	2	1
E	6	6	1

$m_1 = 0.8$

$m_2 = 0.9$

$b = -4.0$

Tasks

1.) Compute Probabilities (5 points)

Customer	Time on site (x_1)	Pages viewed (x_2)	Purchase
A	1	9	0
B	2	3	0
C	3	7	1
D	4.5	2	1
E	6	6	1

Probability
0.1680
0.2315
0.7695
0.6900
0.9608

Customer

A. $z = 0.8(1) + 0.9(9) - 9.0 = -1.6$

B. $z = 0.8(2) + 0.9(3) - 9.0 = -1.2$

C. $z = 0.8(3) + 0.9(7) - 9.0 = 1.2$

D. $z = 0.8(5) + 0.9(2) - 9.0 = 0.8$

E. $z = 0.8(6) + 0.9(6) - 9.0 = 3.2$

2. Computed Average loss

Customer	Time on site (x_1)	(x_2)	(y)
A	1	9	0
B	2	3	0
C	3	7	1
D	5	2	1
E	6	6	1

$\hat{y}$	loss
0.1680	0.1839
0.2315	0.2633
0.7685	0.2633
0.6900	0.3711
0.9600	0.0399

$A = (0 \cdot \ln(0.168) + (1-0) \cdot (1-0.168) = 0.188$

$B = (0 \cdot \ln(0.232) + (1-0)(1-0.332) = 0.263$

$C = (1 \cdot \ln(0.769) + (1-1)(1-0.769) = 0.263$

$D = (1 \cdot \ln(0.690) + (1-1)(1-0.690) = 0.371$

$E = (1 \cdot \ln(0.961) + (1-1)(1-0.961) = 0.040$

$$\text{Loss avg} \; \frac{0.1839 + 0.2633 + 0.2633 + 0.3711 + 0.0394}{5}$$

Loss avg 0.2243

1.1205

5

(3)

Gradient of the loss with respect to b:

$$\frac{\partial L}{\partial b} = \frac{1}{N} \sum (\hat{y} - y_i) = \frac{1}{5}(0.1680 + 0.2315 - 0.2315 - 0.3100 - 0.0342) = -0.03$$

Gradient of the loss with respect to m_1 =

$$\frac{\partial L}{\partial m_1} = \frac{1}{N} \sum (\hat{y} - y_i) x_{1i} = \frac{1}{5}(-1.8477) = -0.3695$$

Gradient of the loss with respect to m_2:

$$\frac{\partial L}{\partial m_2} = \frac{1}{N}\sum(\hat{y}_i - y_i)x_{2i} = \frac{1}{5}(-1.1083) = -0.2217$$

new

$m_1 = 0.8 - 0.1 \times (0.3695) = 0.8370$

$m_2 = 0.9 - 0.1 \times (-0.2217) = 0.9222$

$b = -4.0 - 0.1 \times (-0.0362) = -3.9969$

4. and 5

Customer	Time on Site (x_1)	(x_2)	y	~~$\hat{y}$~~
A	1	4	0	
B	2	3	0	
C	3	7	1	
D	5	2	1	
E	6	6	1	

new $\hat{y}$	new loss
0.1868	0.2068
0.2581	0.2987
0.8131	0.2069
0.7379	0.3046
0.9723	0.0281

$$\text{new loss} = \frac{0.2068 + 0.2987 + 0.2069 + 0.3046 + 0.0281 =}{5}$$

$$\frac{1.0451}{5} = 0.2090$$